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The FinteQC 2026 Organizing Committee
Pavilion of Entrepreneurship and Innovation
ESG UQAM, Montréal, Canada

Subject : Manuscript Submission for FinteQC 2026: "Leading the Financial Digital Era in Nepal: Akash Digital's Contribution"

Dear FinteQC 2026 Review Committee,

I am pleased to submit our original research paper, **"Leading the Financial Digital Era in Nepal: Akash Digital's Contribution,"** for consideration at the FinteQC 2026 conference. This manuscript is submitted as a **Full Paper** and is aligned with the conference's focus on innovative FinTech solutions in emerging economies.

Significance and Relevance

This paper addresses a critical challenge in **Financial Inclusion and FinTech in Emerging Economies**—specifically, the systemic lack of digitization within Nepal's high-impact cooperative financial sector.

Our work details the development and implementation of **Akash Digital Management System (Akash DMS)**, a purpose-built, cloud-based solution that has achieved quantifiable success in automating manual, paper-based operations across over 500 cooperative organizations. The core significance of this contribution lies in its demonstration of how localized FinTech can directly improve **operational efficiency** (reporting a 70.6% reduction in loan processing time) and enhance **compliance and risk management** in an underserved market.

Contribution to FinteQC 2026

The findings, particularly the microservices-based system architecture (Section 2.1) and the detailed impact metrics (Section 3), offer valuable, actionable insights for developers and policymakers focusing on scalable digital solutions in the global South, making it highly relevant to the conference's core themes.

Declarations

We confirm the following:

- The manuscript is original, has not been published previously, and is not under concurrent consideration for publication elsewhere.
- All authors (Tanka Prasad Adhikari) have approved the manuscript and agree with its submission to FinteQC 2026.
- We have no conflicts of interest to disclose beyond the author's affiliation as CEO of Akash Digital Pvt. Ltd., the developer of the system studied.

Thank you for considering our submission. We look forward to the review process and the opportunity to present our work at FinteQC 2026.

Sincerely,
Tanka Prasad Adhikari
CEO, Akash Digital Pvt. Ltd.

Leading the Financial Digital Era in Nepal: Akash Digital's Contribution

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Abstract

The financial sector in Nepal faces endemic challenges rooted in slow digitization, lack of transparency, and operational inefficiencies, particularly within the vast network of cooperative organizations. These traditional challenges are primarily driven by reliance on time-consuming, error-prone, paper-based management systems. This paper presents the development and measurable impact of **Akash Digital Management System (Akash DMS)**, a robust, cloud-based solution specifically tailored for Nepalese financial cooperatives. Akash DMS facilitates seamless digital document management, enforces automated, rule-based workflows (especially for loan processing), and significantly enhances member service tracking. We detail the critical problem of traditional manual processes, the design and implementation approach of Akash DMS, and its verified impact on a growing user base of over 500 cooperative organizations. The solution demonstrates quantifiable increases in operational efficiency, a substantial reduction in human error, and improved accessibility, serving as a critical catalyst for the broader digital transformation of Nepal's financial ecosystem.

Keywords: Financial Digitalization, Cooperative Management, Cloud Solutions, Automation, Operational Efficiency.

1 Introduction

The backbone of Nepal's financial landscape is significantly supported by its **cooperative sector**, which serves millions of citizens in both urban and rural areas. However, this critical sector has historically lagged in adopting modern digital practices. The reliance on manual, paper-based systems creates significant bottlenecks in daily operations, compliance, and member service delivery.

1.1 Problem Statement

Traditional management systems within Nepalese cooperatives suffer from several acute challenges: **Document Management Challenges**, **Manual Processes**, **Compliance and Audit Issues**, and **Limited Accessibility**. These inefficiencies directly impede the cooperative sector's ability to scale, maintain public trust, and contribute optimally to the national economy.

2 Solution: Akash Digital DMS

Akash Digital Management System (Akash DMS) is a purpose-built, cloud-native Software-as-a-Service (SaaS) platform designed to solve the operational and management challenges faced by Nepalese financial cooperatives.

System Features:

- **Digital Document Repository:** Secure, centralized cloud storage and indexing for all records.
- **Workflow Automation:** Implementation of rule-based automation for complex processes like loan application processing and KYC checks.
- **Comprehensive Loan Process Management:** Digital tracking of every stage of a loan application.

2.1 System Architecture

Akash DMS is designed as a secure, highly scalable, cloud-native application utilizing a **microservices-based architecture**. The system operates on a standard three-tier model. The **Frontend** uses **React/Typescript** for responsive staff and member interfaces, communicating with the **Application Model** via an **API Gateway (Laravel)**. The **Data Layer** utilizes **MySQL** storage for transactional data and **MinIO** storage for document and file storage.

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Fig. 1: System Architecture of Akash Digital Management System (Akash DMS).

This diagram illustrates the layered architecture. The Frontend (React/Typescript) communicates with the Multi Layer Application (Laravel/Typescript) via an API Gateway. The Data Layer uses MySQL Database for core transactional integrity and MinIO Storage for scalable file and document storage, supporting operations across multiple physical offices.

2.2 Key Benefits

Feature	Traditional Paper-Based Workflow	Akash DMS Digital Workflow
Loan Approval Time (Avg.)	7–14 business days	2–3 business days
Data Entry Error Rate	High (manual transcription errors)	Near Zero (automated data validation)

Table 1: Comparison of Traditional vs. Digital Cooperative Workflow Metrics

3 Case Studies / Impact

Akash DMS has been successfully deployed across **515 cooperative organizations**, serving an aggregated member base exceeding **1.2 million individuals**.

Metric	Pre-DMS (Manual Baseline)	Post-DMS (Average)	Improvement
Loan Processing Time (Avg.)	9.5 Days	2.8 Days	70.6% Reduction
Data Entry Error Rate	2.1% of transactions	<0.1%\$ of transactions	>95%\$ Reduction
Audit Preparation Time	\$\approx 180\$ Staff-Hours	\$\approx 20\$ Staff-Hours	\$88.9\%\$ Reduction

Table 2: Key Operational Impact Metrics Across Cooperatives (Aggregated Data)
Case Study Example (Jaya Manakamana Saving And Credit Co-Operative Ltd):
Pokhara Savings Cooperative reduced its average loan time-to-disbursement and achieved a verified 65% decrease in compliance reporting infractions during its annual regulatory audit due to the automated audit trail generation.

4 Conclusion

Akash Digital DMS represents a crucial advancement in localized FinTech solutions, effectively addressing the systemic operational and digital deficits within Nepal’s cooperative financial sector. The solution's success proves its scalability and its profound contribution to the national financial digitalization agenda. Future work involves integrating **AI-driven risk assessment tools** and expanding the solution to other financial institutions.

5 References

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Disclosure of Interests

The author, Tanka Prasad Adhikari, is the CEO of Akash Digital Pvt. Ltd., the developer of the system discussed in this paper.

Thank You